

HUMBER ENGINEERING

MENG 3020

SYSTEMS MODELING & SIMULATION

LECTURE 6

LECTURE 6

Rotational Mechanical Systems

- Modeling of Rotational Mechanical Systems
- Variables & Elements
- Element Laws
- Interconnection Laws
- Gear Systems
- Mathematical Modeling of Simple Mechanical Systems

Modeling of Mechanical Systems

- The motion of elements of mechanical systems can be described as:
 - **Translational Motion**
 - **Rotational Motion**
- The equations governing the motion of mechanical systems are called the **equation of motion** that often directly or indirectly formulated by applying **Newton's law of motion** to the **free-body diagram** (FBD).

Translational Motion

$$\sum F_{ext} = Ma$$

$$\sum \tau_{ext} = J\alpha$$

Rotational Motion

- The number of equations of motion required is equal to the number of **linearly independent** motions or the number of **degrees of freedom**.
 - **Step 1:** Identify **reference point** and **positive direction** of motion.
 - **Step 2:** Draw a **free-body diagram** for each inertia or junction with unknown motion.
 - **Step 3:** For each free-body diagram, find the torques acting on the body due only to its own motion and the torques create by the adjacent motion.
 - **Step 4:** Use **Newton's law** on each body to form the **differential equation of motion**.
 - **Step 5:** Represent the equation of motion in **standard forms**.

Rotational Mechanical Systems: Variables & Elements

- The **rotational motion** is defined as a motion that object **rotates** about an **axis**.
- The **variables** that are used to describe the rotational motion are:
 - $\tau(t)$: Torque (N.m)
 - $\theta(t)$: Angular Displacement (rad)
 - $\omega(t)$: Angular Velocity (rad/s)
 - $\alpha(t)$: Angular Acceleration (rad/s²)
- All these variables are function of time.
- **Angular Displacements** are measured with respect to reference angle, which is the equilibrium orientation of the body.
- **Velocities** and **accelerations** are normally expressed as the derivatives of the corresponding **displacement**.
- **Torque** is defined as any causes that tends to produce a change in the rotational motion of a body on which it acts.
- Torque is the **product** of a **force** and the **perpendicular distance** from a point of rotation to the line of action of the force.
- The **elements** that we include in rotational systems are:
 - Rotational Inertia Element → **Moment of Inertia**
 - Rotational Stiffness Element → **Torsional Spring**
 - Rotational Friction Element → **Rotational Damper**

$$\omega(t) = \frac{d\theta(t)}{dt} = \theta'(t) = \dot{\theta}(t)$$

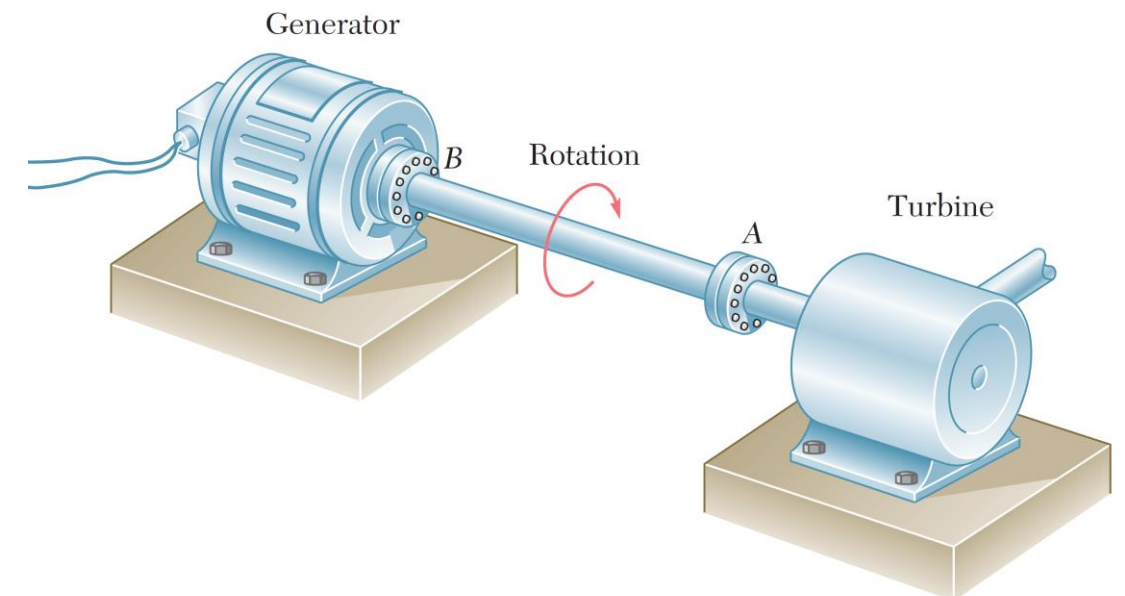
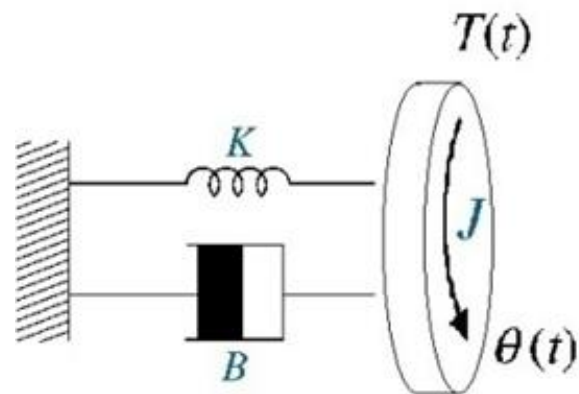
$$\alpha(t) = \frac{d^2\theta(t)}{dt^2} = \theta''(t) = \ddot{\theta}(t)$$

Rotational Mechanical Systems: Example

- Some real-world examples of rotational motion systems.

○ Transmission Shaft

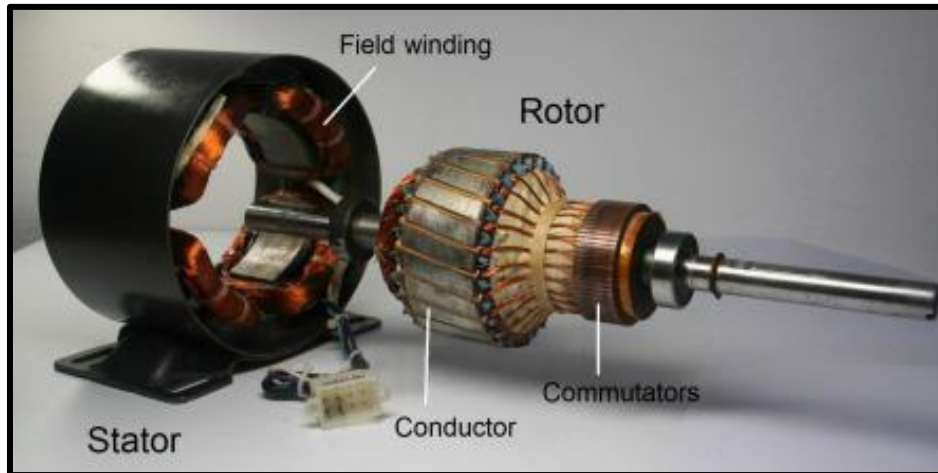
- The **transmission shaft** is a common component of the rotational systems to **transmit the power** from one point to another. It is also a practical example of a rotational inertia-spring-damper system.
- The shaft (rod) under torsional load can be modeled as a rotational **inertia-spring-damping** system.
- The inertia of the rod is represented by **inertia disk J** .
- The compliance and flexibility of a rod or a shaft when it is subject to an applied torque can be modeled by a **torsional spring K** .
- The internal energy loss in a rod is represented by **viscus damping B** .



Rotational Mechanical Systems: Example

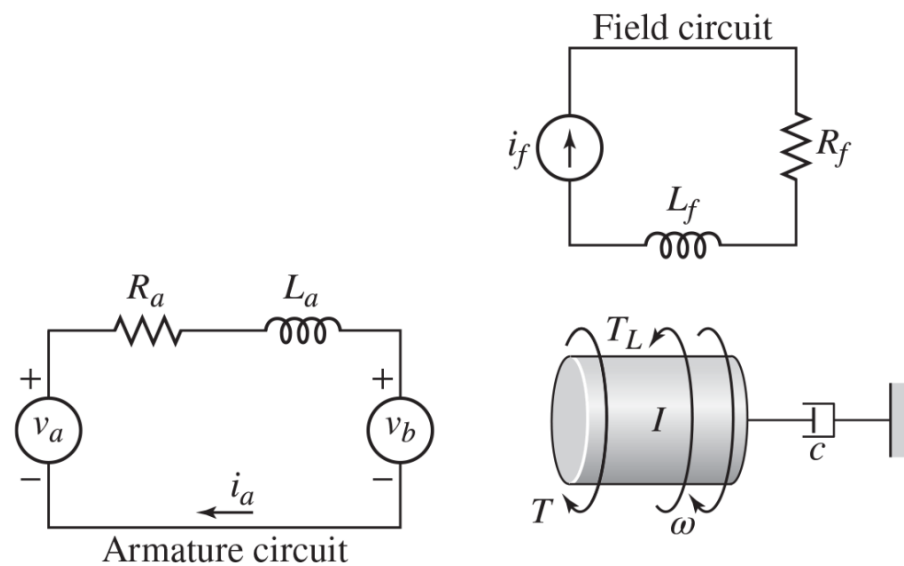
- Some real-world examples of rotational motion systems.

○ Armature Controlled DC Motor

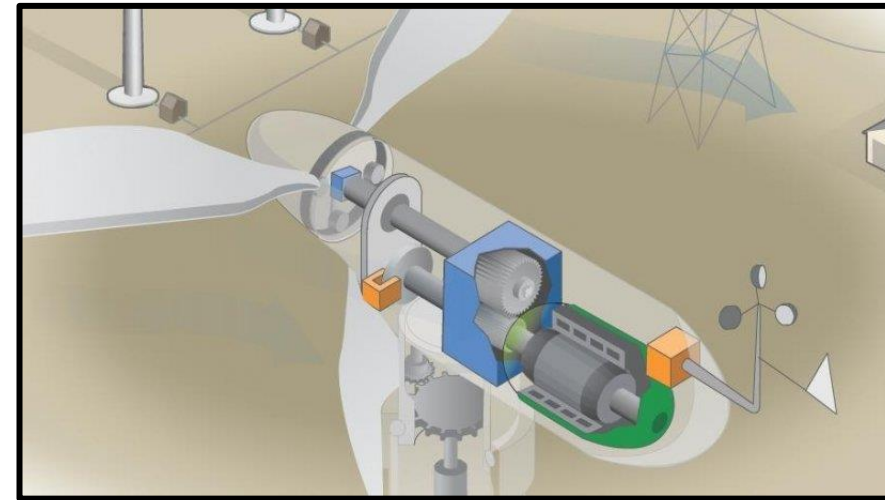


Input: Applied armature voltage

Output: Angular speed of rotor shaft

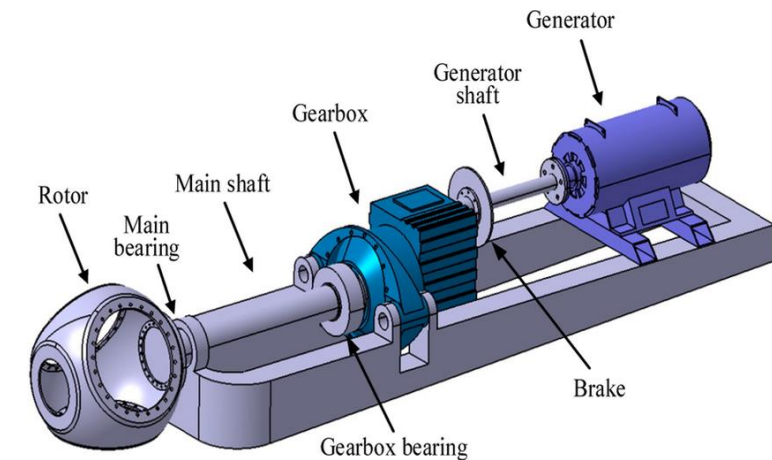
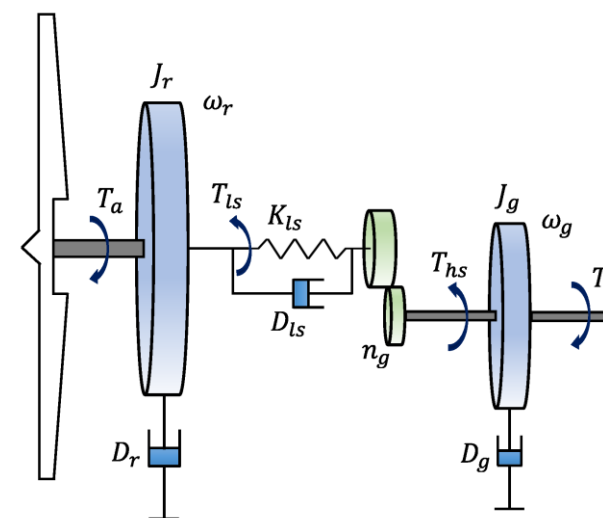


○ Wind Turbine Drivetrain



Input: Applied torque

Output: Speed of generator

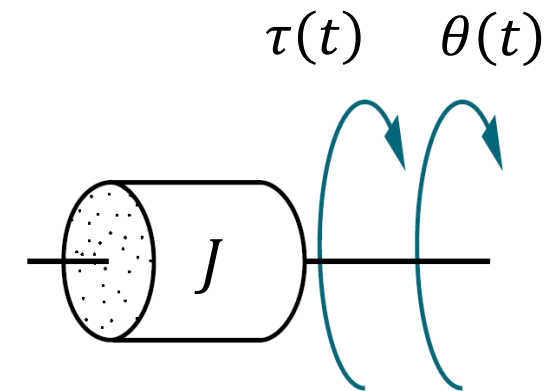


Rotational Mechanical Systems: Element Laws

□ Inertia Element: Moment of Inertia

- If a **torque** is applied on a body having **inertia**, then it is opposed by an opposing torque due to the **moment of inertia**.
- Physically, the **moment of inertia** of a body is a measure of the **resistance** of the body to **angular acceleration**.
- From the **Newton's second law**, the applied **torque** is proportional to **angular acceleration** of the body.
- The J is the **moment of inertia**. The unit is (kg.m^2) .

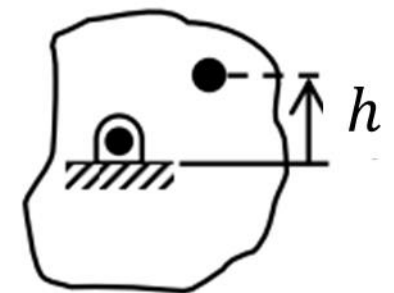
$$\tau(t) = J\alpha(t) = J \frac{d\omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$$



- **Energy** in a rotating body can store in both **kinetic energy** and **potential energy** forms.

$$KE = \frac{1}{2}J\omega^2$$

$$PE = Mgh$$

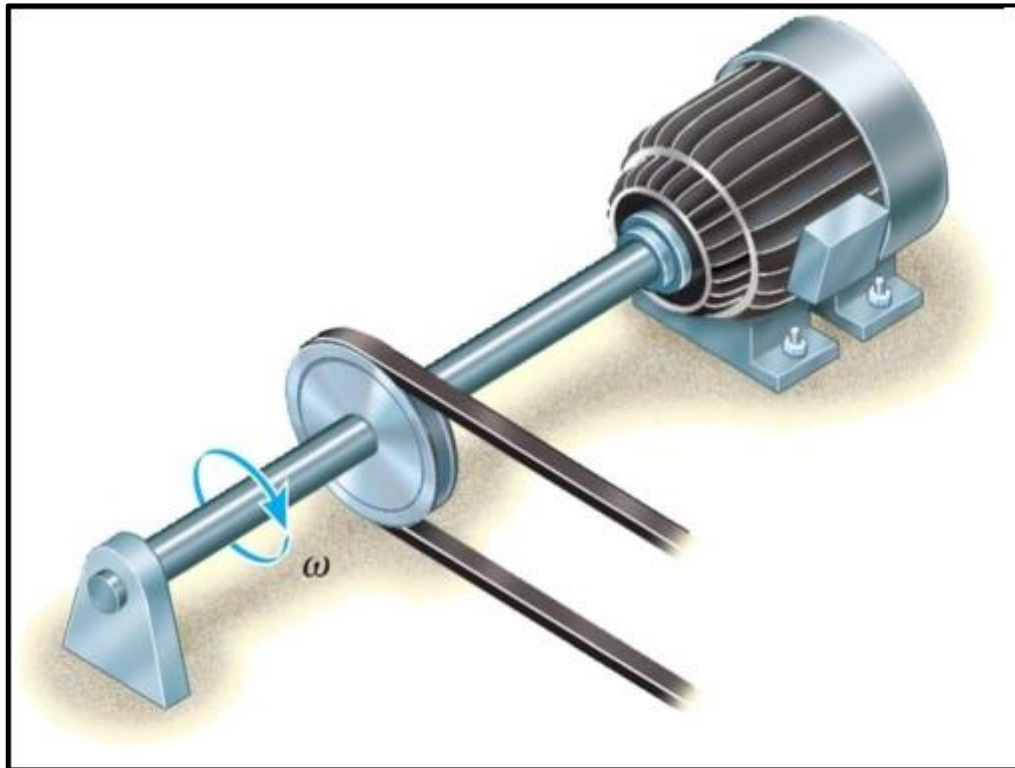


where M is the mass, $g = 9.81 \text{ m/s}^2$ is **gravitational acceleration** and h is the **height of the center of mass above its reference position**.

Rotational Mechanical Systems: Element Laws

□ Rotational Stiffness Element

- Any mechanical element that undergoes a change in shape when subjected to a **force** or **torque** can be characterized by a **stiffness element**.
- The most common rotational stiffness elements are the **torsional springs** and a **rotational shafts**.

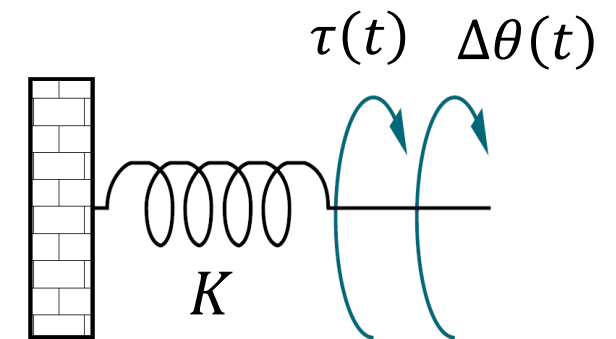


Rotational Mechanical Systems: Element Laws

□ Rotational Stiffness Element

- If a **torque** is applied on a torsional spring or a shaft, then it is opposed by an opposing torque due to the **elasticity** of the element.
- The applied **torque** $\tau(t)$ is proportional to the **angular displacement** $\Delta\theta(t)$ of the torsional spring or the shaft.
- The K is the **stiffness constant** or **torsional spring constant**. The unit is (N.m/rad).

$$\tau(t) = K\Delta\theta(t)$$



- **Potential energy** is stored in a **twisted stiffness element** and can affect the response of the system at later time.
For a linear torsional spring or a shaft, the potential energy is:

$$PE = \frac{1}{2}K(\Delta\theta)^2$$

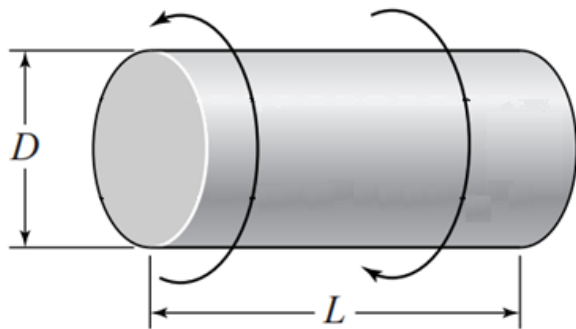
- All practical stiffness elements have **inertia** and **damping**.
- Here we assume that the effect of the inertia and damping effects are negligibly small.

Rotational Mechanical Systems: Element Laws

□ Stiffness Constant / Torsional Spring Constant

- Torsional spring constants depend on the **geometry** of the element and its **material properties**.

Solid shaft



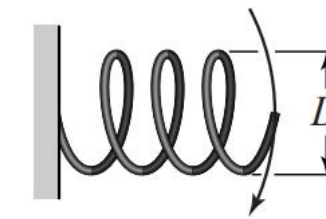
$$K = \frac{\pi G D^4}{32 L}$$

D = shaft diameter

L = shaft length

G = Shear modulus of elasticity

Coil spring



$$K = \frac{E d^4}{64 n D}$$

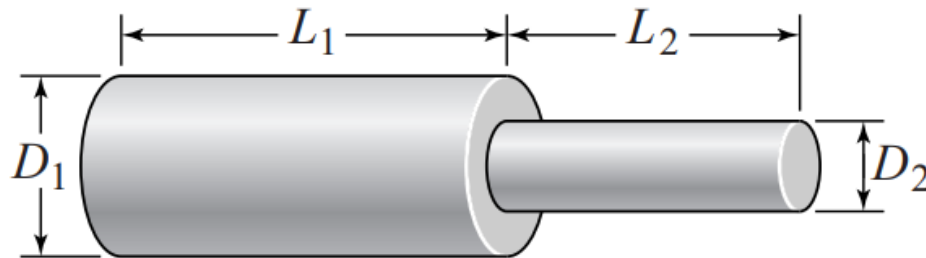
D = coil diameter

n = number of coils

d = wire diameter

E = Shear modulus of elasticity

- The equivalent torsional spring constant for the **stepped shaft** is:



$$\frac{1}{K_e} = \frac{1}{K_1} + \frac{1}{K_2} \quad \rightarrow \quad K_e = \frac{K_1 K_2}{K_1 + K_2}$$

Rotational Mechanical Systems: Element Laws

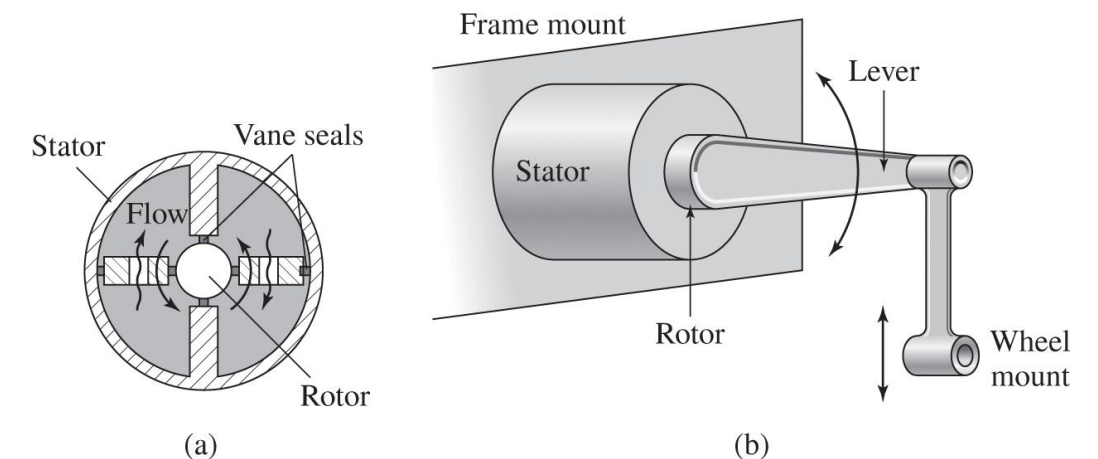
□ Rotational Friction Element

- Rotational viscous friction arises when two rotating bodies are separated by a film of oil.
- For example, the door closer rotary dampers or shock absorber dampers, ball bearings, clutches, torque limiters and brakes (mechanical friction).

Torque limiters & Brakes



Rotary damper



Ball bearings

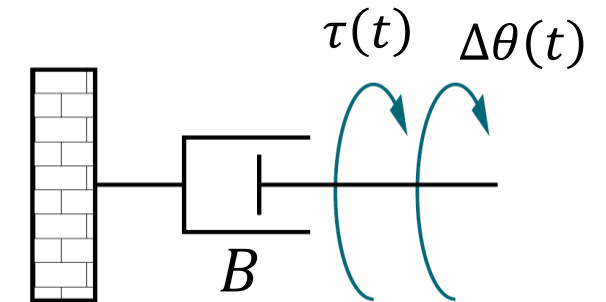


Rotational Mechanical Systems: Element Laws

□ Rotational Friction Element

- A rotary damper or torsional damper exerts a resisting torque $\tau(t)$ in response to an angular velocity difference $\Delta\omega(t)$ across it.
- The torque due to friction is proportional to the angular velocity of the motion, and its direction tends to reduce the relative angular velocity.
- The B is the viscous friction coefficient. The unit is (N.m.s/rad)

$$\tau(t) = B\Delta\omega(t) = B \frac{d\Delta\theta(t)}{dt}$$



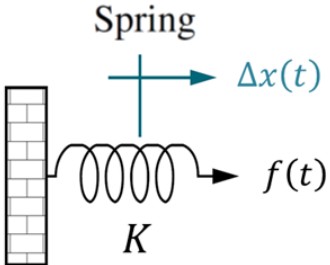
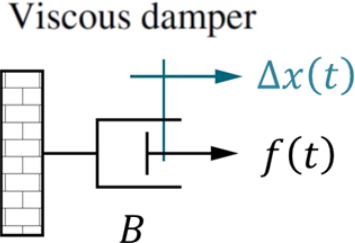
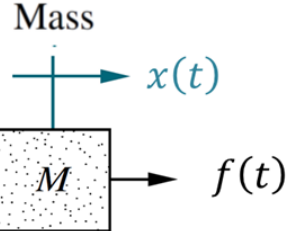
- All practical dampers produce inertia and spring effects. Here, we assume that these effects are negligible.

Mechanical Systems: Element Laws

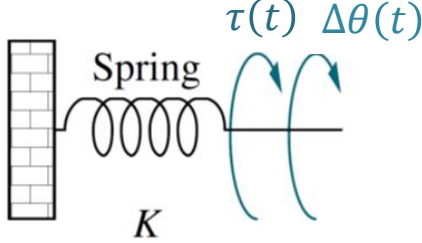
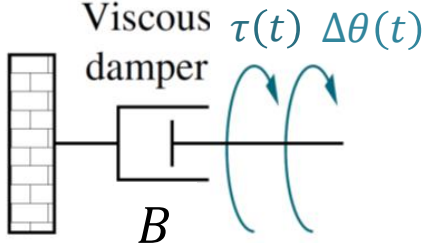
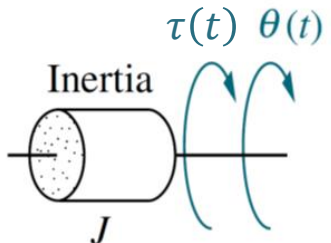
□ Summary

- Table shows summary of the **Input-Output** relationships for translational and rotational motions.

Translational Motion

Element	Force-velocity	Force-displacement
	$f(t) = K \int_0^t v(t) dt$	$f(t) = K \Delta x(t)$
	$f(t) = B \Delta v(t)$	$f(t) = B \frac{d\Delta x(t)}{dt}$
	$f(t) = M \frac{dv(t)}{dt}$	$f(t) = M \frac{d^2 x(t)}{dt^2}$

Rotational Motion

Element	Torque-angular velocity	Torque-angular displacement
	$\tau(t) = K \int_0^t \omega(t) dt$	$\tau(t) = K \Delta \theta(t)$
	$\tau(t) = B \Delta \omega(t)$	$\tau(t) = B \frac{d\Delta \theta(t)}{dt}$
	$\tau(t) = J \frac{d\omega(t)}{dt}$	$\tau(t) = J \frac{d^2 \theta(t)}{dt^2}$

Rotational Mechanical Systems: Interconnection Laws

□ Newton's Second Law

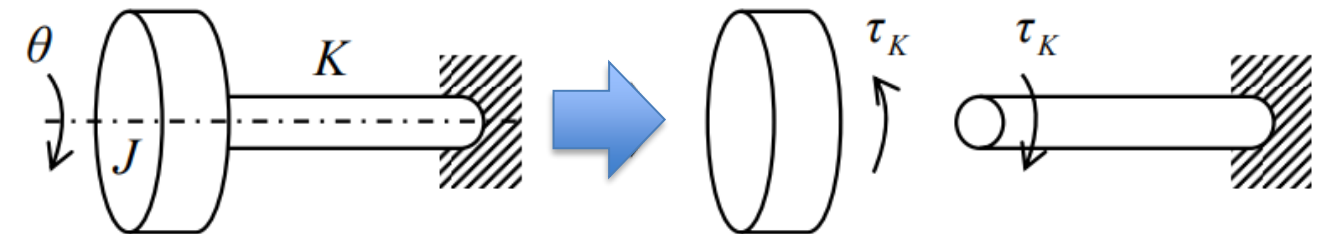
- For a **rigid body** in pure rotation about a fixed axis, if $\sum \tau_{ext}$ is the sum of all torques acting about a given axis and J is the moment of inertia of a body about that axis, then,

$$\sum \tau_{ext} = J\alpha(t)$$

where α is the angular acceleration of the body.

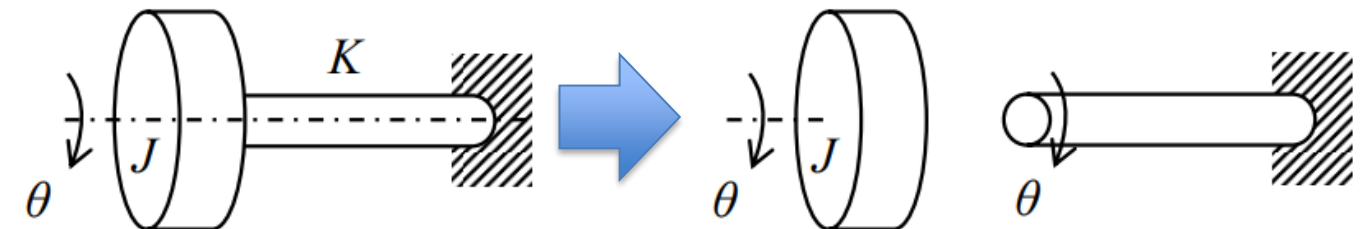
□ Newton's Third Law: The Law of Reaction Torques

- For bodies that are rotating about the same axis, any torque exerted by one element on another is accompanied by a **reaction torque** of equal magnitude and opposite direction on the first element.
- Elements must be connected along the **same axis**.
(Not applicable for gear meshing)



□ The Law for Angular Displacement

- Two elements connected along the same axis have the **same angular displacement**. (Not applicable to gear meshing since the axes are not the same)



Modeling of Rotational Mechanical Systems

Example 1

Consider the following rotational system.

a) Find the equation of motion and the transfer function model.

Assume that the input is the applied torque τ_a and the output is the angular displacement of the disk θ .

From the **free-body diagram** by considering the **CW** direction as the **positive direction**, the **equation of motion** is obtained.

$$\sum \tau_{ext} = J\alpha(t) \rightarrow \tau_a(t) - \tau_K(t) - \tau_B(t) = J\alpha(t)$$

$$\tau_a(t) - K\theta(t) - B\dot{\theta}(t) = J\ddot{\theta}(t) \rightarrow \tau_a(t) = J\ddot{\theta}(t) + B\dot{\theta}(t) + K\theta(t)$$

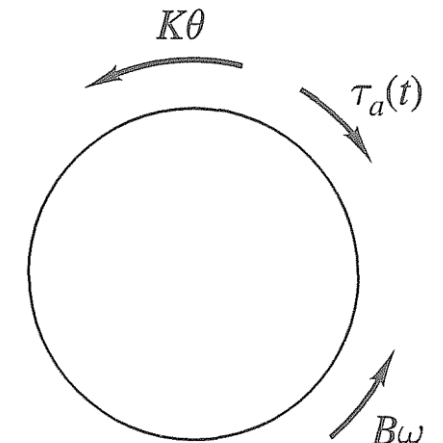
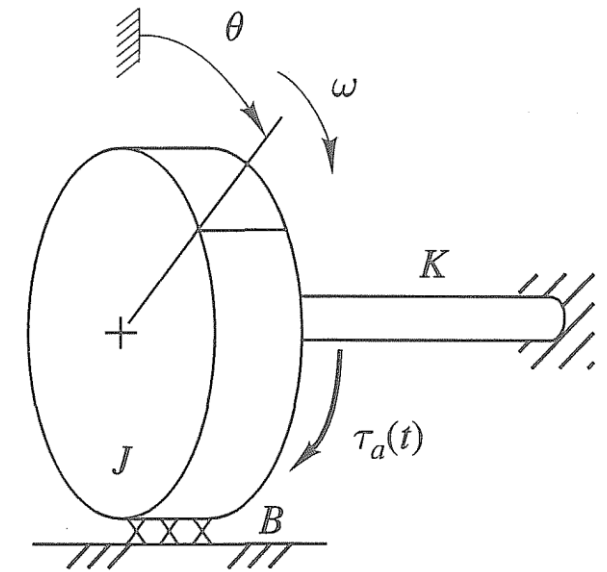
Equation of Motion

Taking the **Laplace transform**, assuming **zero initial conditions**, and solving for the transfer function

$$T_a(s) = Js^2\theta(s) + Bs\theta(s) + K\theta(s) \rightarrow T_a(s) = (Js^2 + Bs + K)\theta(s)$$

$$\frac{\theta(s)}{T_a(s)} = \frac{1}{Js^2 + Bs + K}$$

Transfer Function Model



Modeling of Rotational Mechanical Systems

Example 1

Consider the following rotational system.

b) Derive a state-space model for the following rotational system.

Assume that the input is the applied torque τ_a and two outputs are the angular acceleration of the disk α , and the torque exerted on the disk by the shaft τ_K .

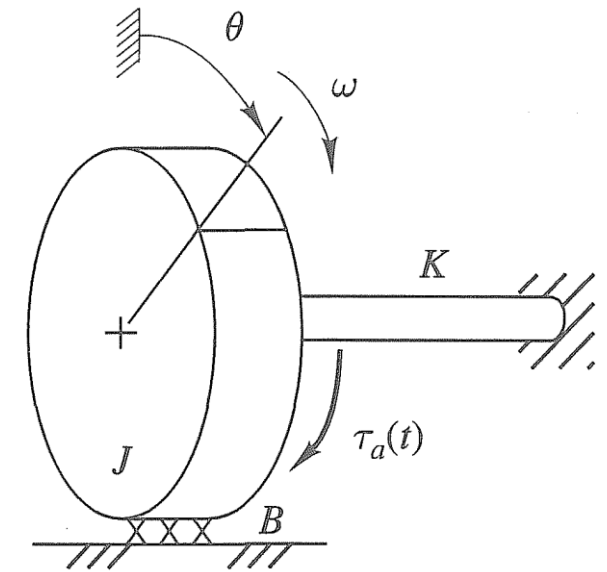
Having the equation of motion: $\tau_a(t) = J\ddot{\theta}(t) + B\dot{\theta}(t) + K\theta(t)$

Define the state variables q_1 and q_2 as the angular displacement of the shaft (spring K) and the angular velocity of the disk (inertia J).

$$q_1 = \theta \quad \rightarrow \quad \dot{q}_1 = \dot{\theta} \quad \rightarrow \quad \dot{q}_1 = q_2 \quad \text{Eqn. (1)}$$

$$q_2 = \dot{\theta} \quad \rightarrow \quad \dot{q}_2 = \ddot{\theta} \quad \rightarrow \quad \dot{q}_2 = \frac{1}{J}\tau_a(t) - \frac{K}{J}\theta - \frac{B}{J}\dot{\theta}$$

$$\rightarrow \quad \dot{q}_2 = \frac{1}{J}\tau_a(t) - \frac{K}{J}q_1 - \frac{B}{J}q_2 \quad \text{Eqn. (2)}$$



$$\text{outputs} \rightarrow \begin{cases} y_1 = \alpha & \rightarrow & y_1 = \frac{1}{J}\tau_a(t) - \frac{K}{J}q_1 - \frac{B}{J}q_2 \\ y_2 = \tau_K & \rightarrow & y_2 = K\theta = Kq_1 \end{cases}$$

Modeling of Rotational Mechanical Systems

Example 1

Consider the following rotational system.

b) Derive a state-space model for the following rotational system.

Assume that the input is the applied torque τ_a and two outputs are the angular acceleration of the disk α , and the torque exerted on the disk by the shaft τ_K .

From the state equations and the output equations:

$$\begin{cases} \dot{q}_1 = q_2 \\ \dot{q}_2 = \frac{1}{J} \tau_a(t) - \frac{K}{J} q_1 - \frac{B}{J} q_2 \end{cases} \quad \begin{cases} y_1 = \frac{1}{J} \tau_a(t) - \frac{K}{J} q_1 - \frac{B}{J} q_2 \\ y_2 = K q_1 \end{cases}$$

We can represent the state and output equations in the standard matrix-vector form as below:

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)$$

State Equation



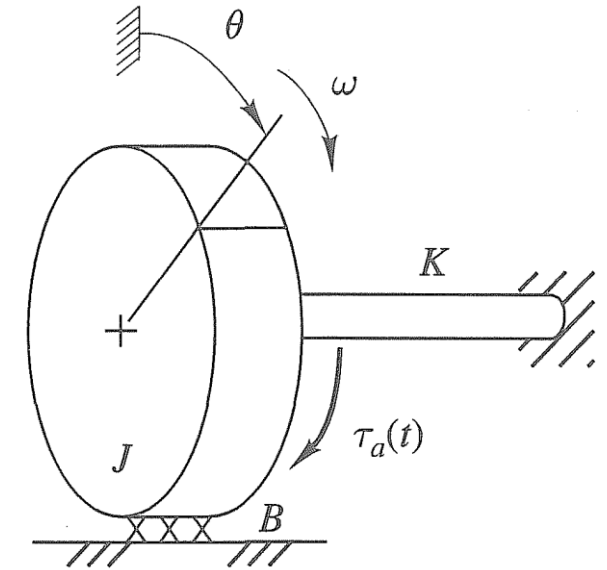
$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{K}{J} & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix} \tau_a(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}u(t)$$

Output Equation



$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{K}{J} & -\frac{B}{J} \\ K & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{J} \\ 0 \end{bmatrix} \tau_a(t)$$



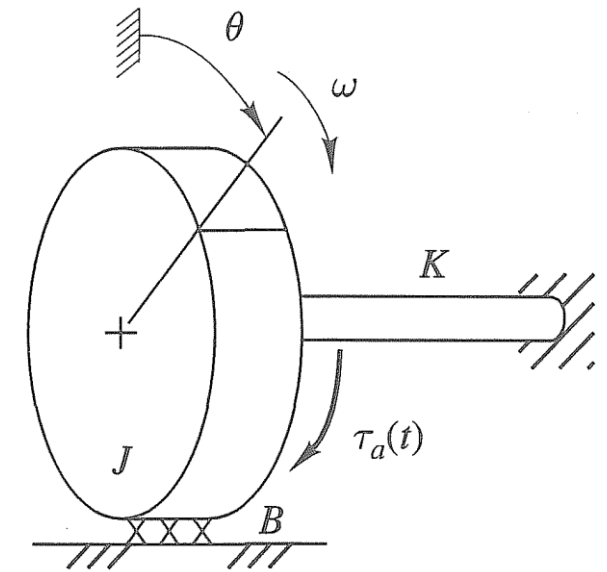
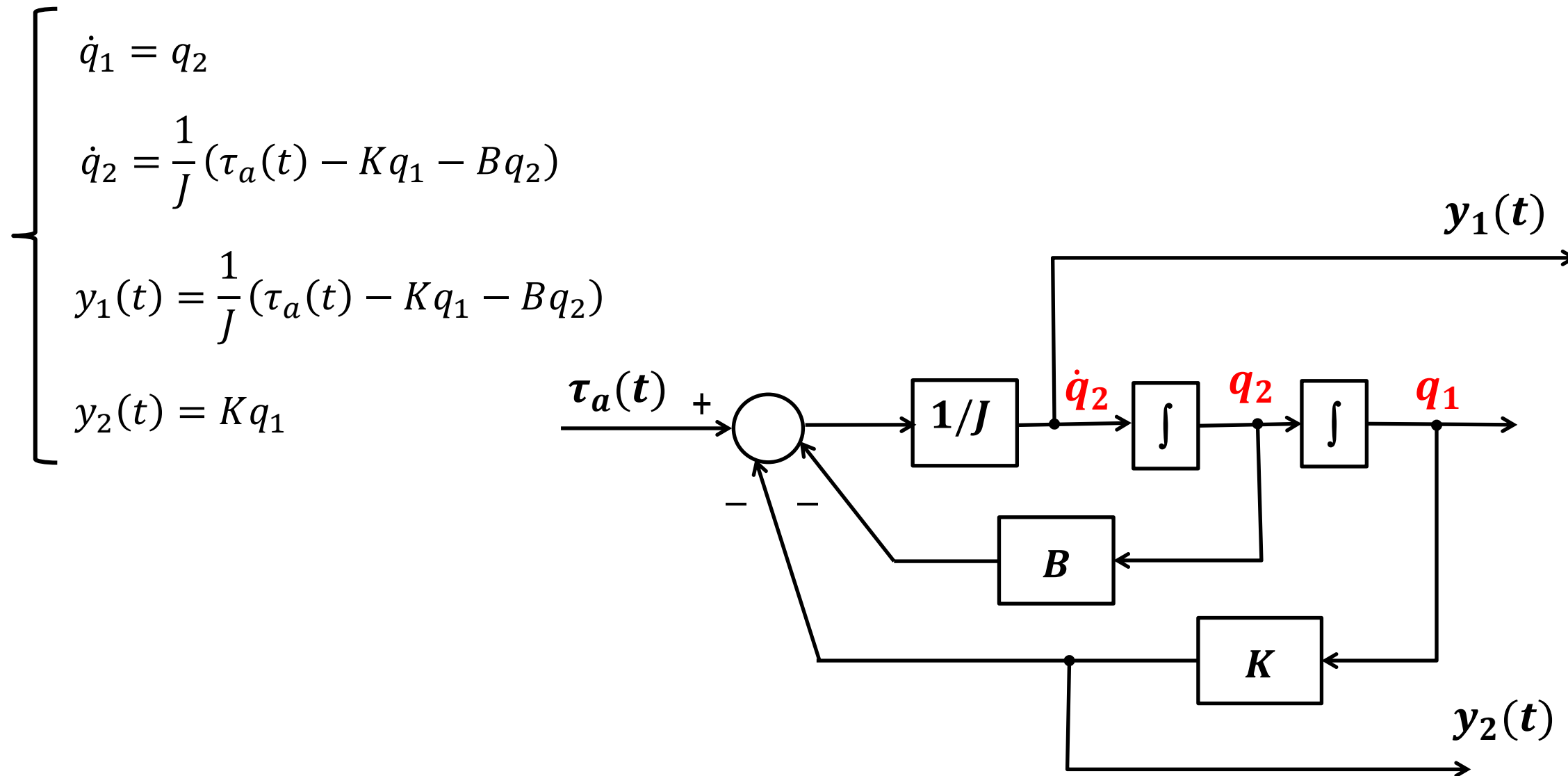
Modeling of Rotational Mechanical Systems

Example 1

Consider the following rotational system.

c) Draw the block diagram model of the system using the state-space model.

Following block diagram visualizes the state variables and the system outputs.

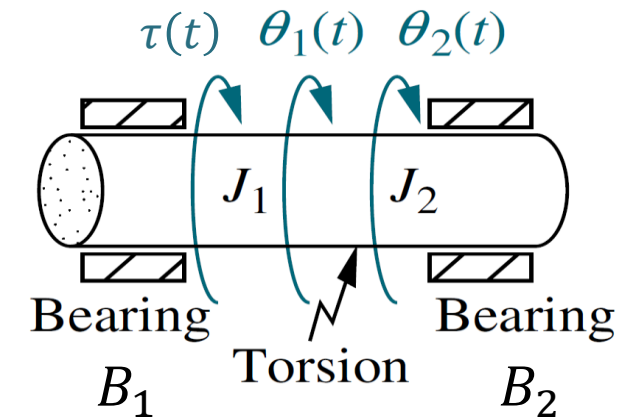
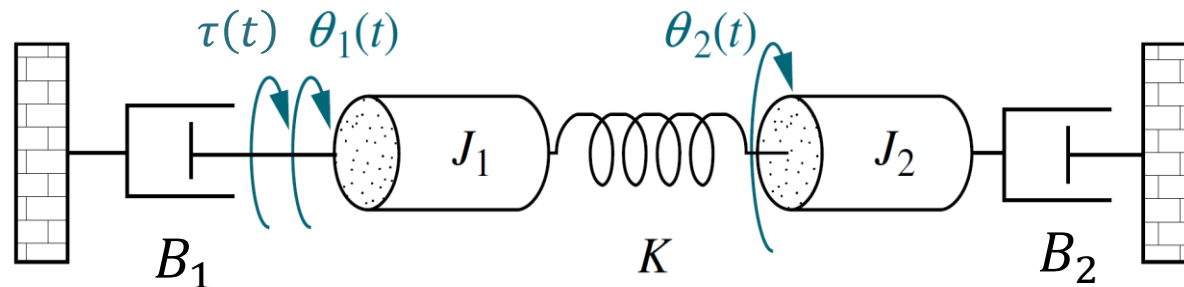


Modeling of Rotational Mechanical Systems

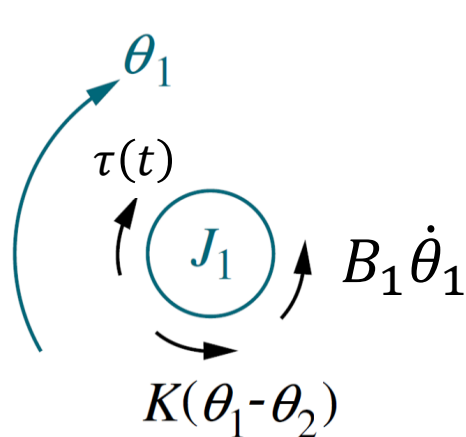
Example 2

Find the equation of motion and a transfer function model of $\theta_2(s)/T(s)$, for the following rotational system. The rod is supported by bearings at either end and is undergoing torsion. A torque is applied at the left, and the displacement is measured at the right.

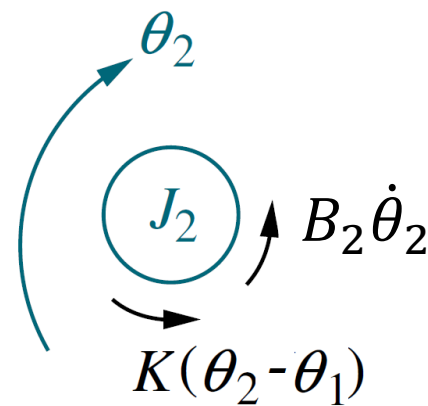
The system can be modeled as an **inertia-spring-damper** system as below



Draw the **free-body diagram** by considering the **CW** as the **positive direction**.



Inertia J_1



Inertia J_2



Modeling of Rotational Mechanical Systems

Example 2

Find the equation of motion and a transfer function model of $\theta_2(s)/T(s)$, for the following rotational system. The rod is supported by bearings at either end and is undergoing torsion. A torque is applied at the left, and the displacement is measured at the right.

The equation of motion for each rotational inertia is obtained:

$$\text{Inertia } J_1 \rightarrow \tau(t) - K(\theta_1 - \theta_2) - B_1\dot{\theta}_1 = J_1\ddot{\theta}_1$$

$$\text{Inertia } J_2 \rightarrow -K(\theta_2 - \theta_1) - B_2\dot{\theta}_2 = J_2\ddot{\theta}_2$$

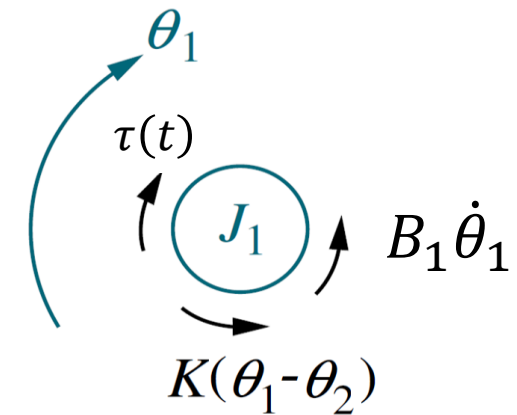
Take the Laplace transform, assuming zero initial conditions, and solve for the transfer function

$$T(s) - K\theta_1(s) + K\theta_2(s) - B_1s\theta_1(s) = J_1s^2\theta_1(s)$$

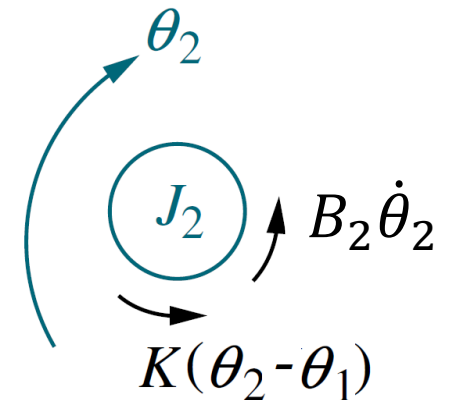
$$T(s) = (J_1s^2 + B_1s + K)\theta_1(s) - K\theta_2(s) \rightarrow \text{Eqn. (1)}$$

$$-K\theta_2(s) + K\theta_1(s) - B_2s\theta_2(s) = J_2s^2\theta_2(s)$$

$$K\theta_1(s) - (J_2s^2 + B_2s + K)\theta_2(s) = 0 \rightarrow \text{Eqn. (2)}$$



Inertia J_1



Inertia J_2

Modeling of Rotational Mechanical Systems

Example 2

Find the equation of motion and a transfer function model of $\theta_2(s)/T(s)$, for the following rotational system. The rod is supported by bearings at either end and is undergoing torsion. A torque is applied at the left, and the displacement is measured at the right.

$$T(s) = (J_1 s^2 + B_1 s + K)\theta_1(s) - K\theta_2(s) \rightarrow \text{Eqn. (1)}$$

$$K\theta_1(s) - (J_2 s^2 + B_2 s + K)\theta_2(s) = 0 \rightarrow \text{Eqn. (2)}$$

Find $\theta_1(s)$ from Eqn. (2) in terms of $\theta_2(s)$ and substitute in Eqn. (1):

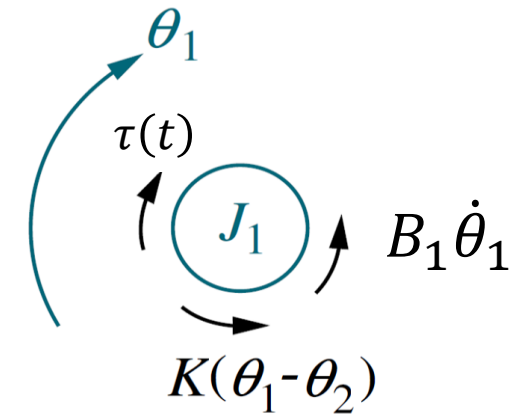
$$\text{From Eqn. (2)} \rightarrow \theta_1(s) = \frac{1}{K}(J_2 s^2 + B_2 s + K)\theta_2(s)$$

$$T(s) = (J_1 s^2 + B_1 s + K) \left(\frac{1}{K}(J_2 s^2 + B_2 s + K)\theta_2(s) \right) - K\theta_2(s)$$

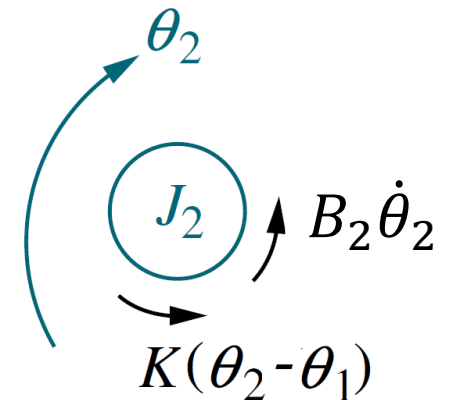
$$KT(s) = (J_1 s^2 + B_1 s + K)(J_2 s^2 + B_2 s + K)\theta_2(s) - K^2\theta_2(s)$$

$$\frac{\theta_2(s)}{T(s)} = \frac{K}{(J_1 s^2 + B_1 s + K)(J_2 s^2 + B_2 s + K) - K^2}$$

Transfer Function Model



Inertia J_1

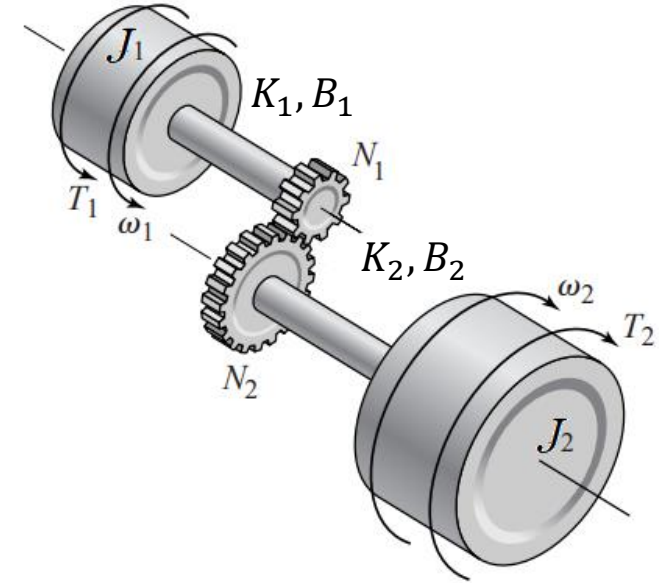


Inertia J_2

Modeling of Rotational Mechanical Systems

□ Rotational Transformer: Gears

- The **gear** is a device that transmits energy from one part of the rotational system to another in such a way that **force**, **torque**, **speed**, and **displacement** may be altered.
- Gears allow us to **match the drive system** and **the load**—a trade-off between **speed** and **torque**.
- They also change **direction** of rotational motion.
- **Ideal gears**, have **no moment of inertia**, **no friction** and **no backlash** (perfect meshing of teeth).
- Actual gears have inertia, friction and backlash, but these can be represented by additional elements.
- The spacing between teeth must be **equal** for each gear in a pair, so the radii of the gears are **proportional** to the number of teeth.

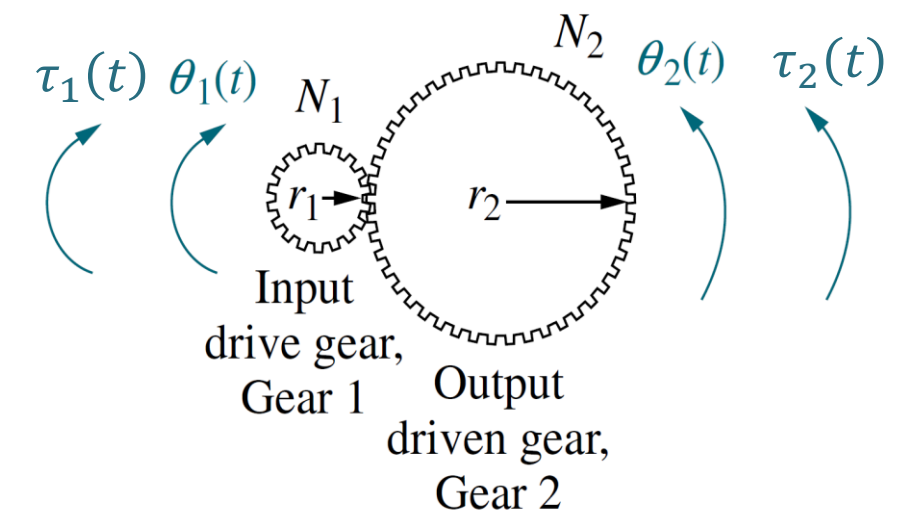
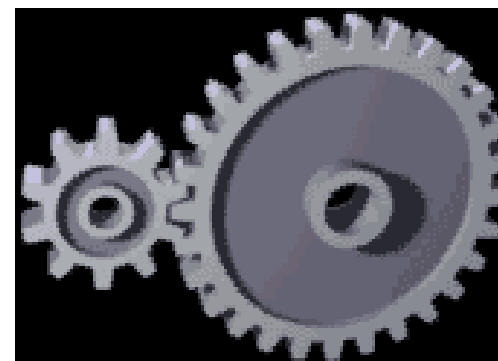


$$\frac{N_2}{N_1} = \frac{r_2}{r_1}$$

Teeth numbers

Radii of the gears

- The N is called the **gear ratio**.
$$N = \frac{N_2}{N_1}$$



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□ Rotational Transformer: Gears

- Assume the **ideal gears** shown below, where points A and B denote points on the circles that are in contact with each other at some reference time.
- At some later time, A and B will have moved to the positions shown, where θ_1 and θ_2 denote the respective displacements from their original position.
- Since the arc PA and PB must be equal,

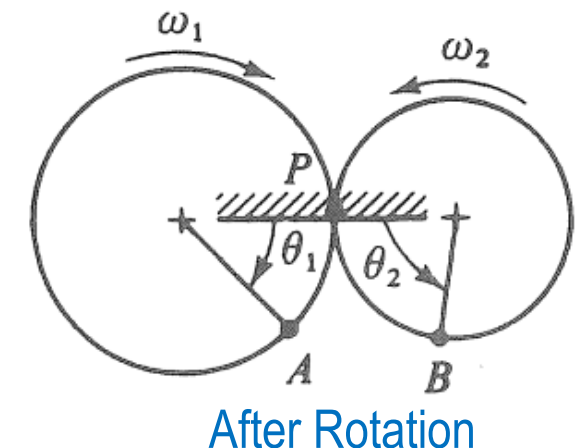
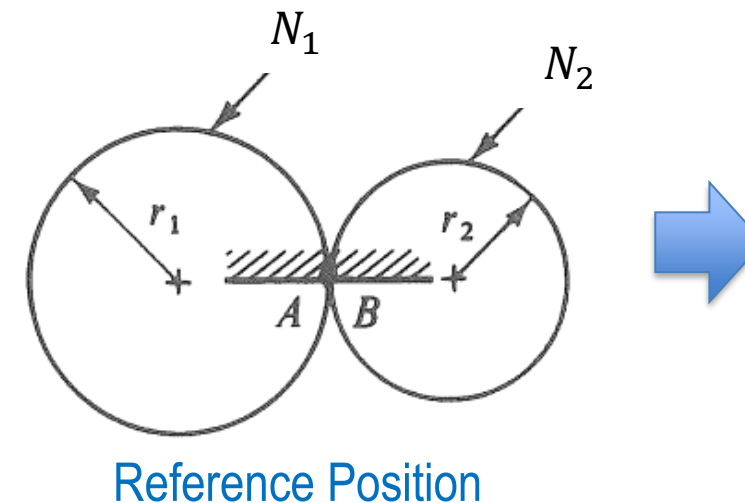
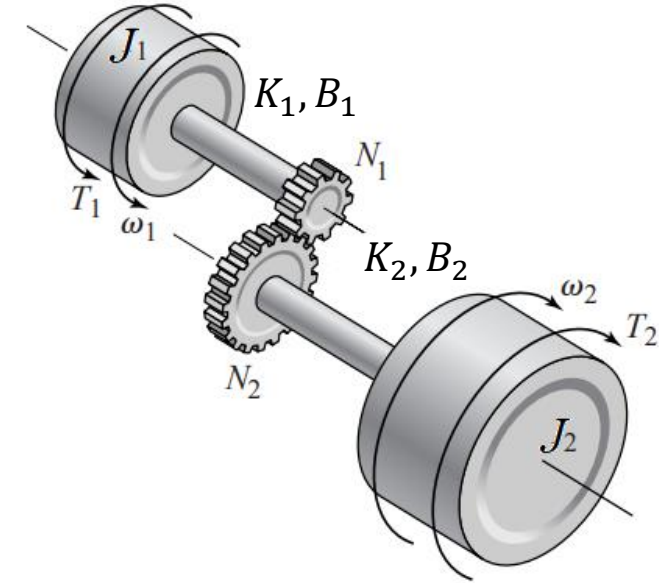
$$r_1 \theta_1 = r_2 \theta_2 \quad \longrightarrow \quad \boxed{\frac{\theta_1}{\theta_2} = \frac{r_2}{r_1}} \quad \longrightarrow \quad \boxed{\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1}}$$

- Ideal gears are **lossless**.
- They do not absorb or store energy, the energy into Gear 1 equals the energy out of Gear 2,

$$\tau_1 \theta_1 = \tau_2 \theta_2 \quad \longrightarrow \quad \boxed{\frac{\theta_1}{\theta_2} = \frac{\tau_2}{\tau_1}}$$

- The **law of gear transformation** for system variables is:

$$\rightarrow \boxed{\frac{\tau_2}{\tau_1} = \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} = \frac{N_2}{N_1} = N}$$



Modeling of Rotational Mechanical Systems

□ Gear Equivalent System

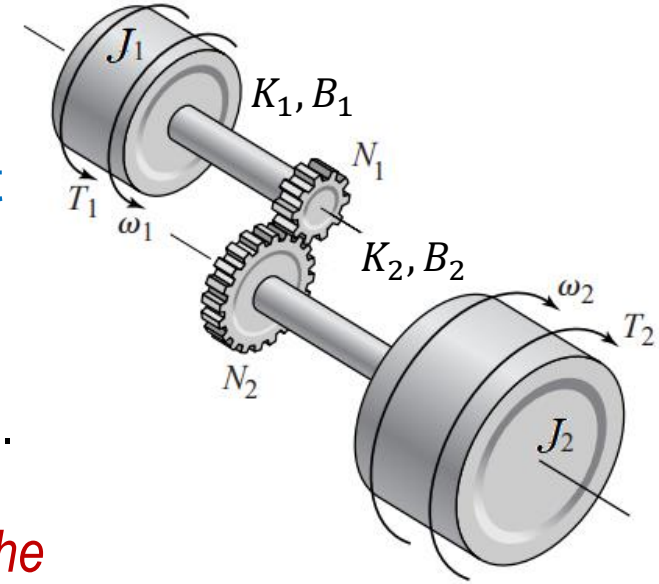
- The **gear equivalent system** refers to the simplification of **complex gear train** into a **single equivalent gear** or a **gearless system** for easier mathematical modeling and analysis.
- The equivalent system is obtained by **reflecting** all components to a single shaft.
- The technique uses the principle of equivalent **impedance reflection** similar to transformers analysis.
- *Rotational mechanical impedances (J , B , K) can be reflected through gear trains by **multiplying the mechanical impedance by the ratio**:*

$$\left(\frac{\text{Number of teeth of gear on destination shaft}}{\text{Number of teeth of gear on source shaft}} \right)^2$$

*where the impedance to be reflected is attached to the **source shaft** and is being reflected to the destination shaft.*

- For example, in the gears shown in this system.
- The expression for the **equivalent impedances** felt on the **shaft 1** are:

$$J_e = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2 \quad \text{and} \quad B_e = B_1 + B_2 \left(\frac{N_1}{N_2} \right)^2 \quad \text{and} \quad K_e = K_1 + K_2 \left(\frac{N_1}{N_2} \right)^2$$



Modeling of Rotational Mechanical Systems

Example 3

The following system shows gears driving a rotational inertia, spring, and viscous damper. Find the transfer function, $\theta_1(s)/T_1(s)$ for the following system.

First, represent the system as an equivalent **gearless** system.

From the gear law, τ_1 can be reflected to the output by multiplying by the **gear ratio** N_2/N_1 .

$$\tau_2 = \tau_1 \left(\frac{N_2}{N_1} \right) = N \tau_1$$

The **equation of motion** is:

$$\tau_2(t) - K\theta_2(t) - B\dot{\theta}_2(t) = J\ddot{\theta}_2(t)$$

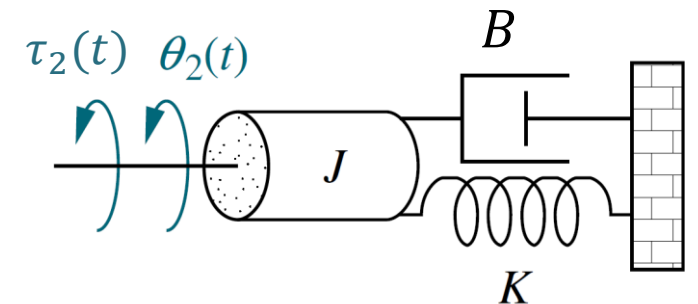
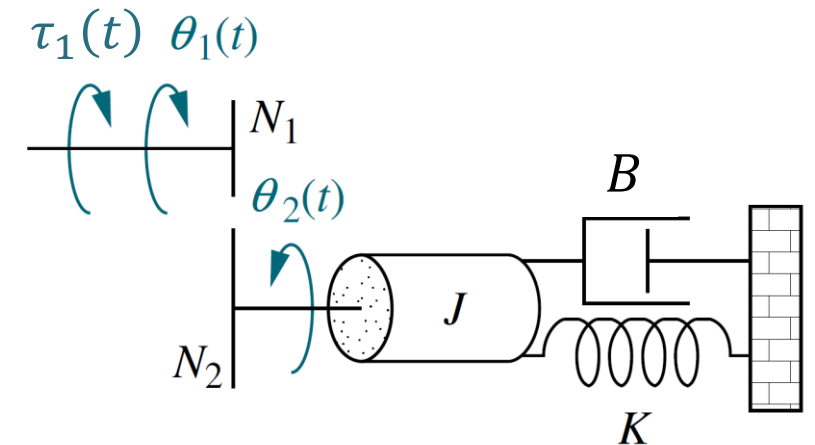
Now convert, θ_2 into an equivalent θ_1 by considering the **gear ratio** effect.

$$\theta_2 = \theta_1 \left(\frac{N_1}{N_2} \right) = \frac{1}{N} \theta_1$$

The **equation of motion** in terms of input $\tau_1(t)$ and output $\theta_1(t)$ is:

$$N\tau_1(t) = K \left(\frac{1}{N} \right) \theta_1(t) + B \left(\frac{1}{N} \right) \dot{\theta}_1(t) + J \left(\frac{1}{N} \right) \ddot{\theta}_1(t)$$

Simplify the equation as $\rightarrow$ $N^2\tau_1(t) = K\theta_1(t) + B\dot{\theta}_1(t) + J\ddot{\theta}_1(t)$



$$\frac{\tau_2}{\tau_1} = \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} = \frac{N_2}{N_1} = N$$

Modeling of Rotational Mechanical Systems

Example 3

The following system shows gears driving a rotational inertia, spring, and viscous damper. Find the transfer function, $\theta_1(s)/T_1(s)$ for the following system.

Having the equation of motion:

$$N^2\tau_1(t) = K\theta_1(t) + B\dot{\theta}_1(t) + J\ddot{\theta}_1(t)$$

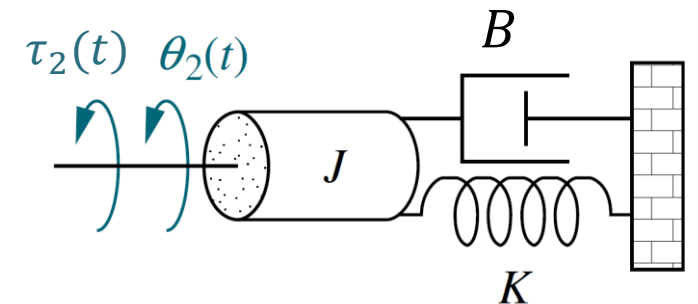
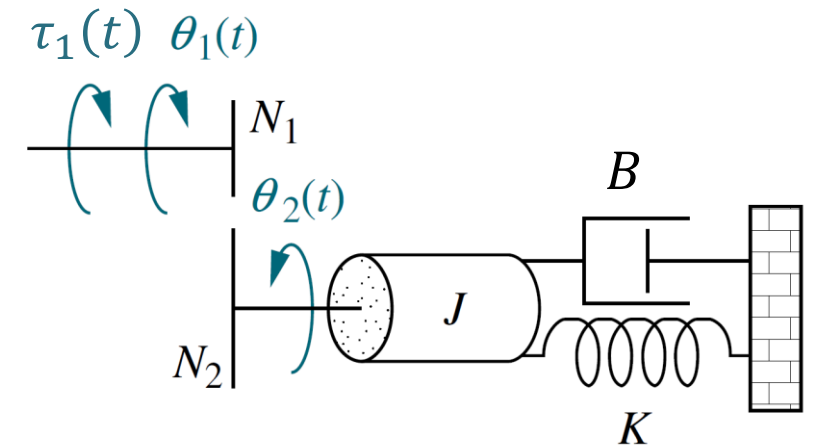
Take Laplace transform with zero initial conditions to find the transfer function:

$$N^2T_1(s) = K\theta_1(s) + Bs\theta_1(s) + Js^2\theta_1(s)$$

$$N^2T_1(s) = (K + Bs + Js^2)\theta_1(s)$$

$$\frac{\theta_1(s)}{T_1(s)} = \frac{N^2}{Js^2 + Bs + K}$$

Transfer Function Model



$$\frac{\tau_2}{\tau_1} = \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} = \frac{N_2}{N_1} = N$$

Modeling of Rotational Mechanical Systems

Example 4

The following figure shows gears driving a rotational inertia, spring, and viscous damper system. Find the transfer function, $\theta_2(s)/T_1(s)$ for the following system.

First, find the equivalent **gearless** system.

From the **gear law** the torque τ_1 can be reflected to the output by multiplying by the **gear ratio** N .

$$\tau_2 = \frac{N_2}{N_1} \tau_1 = N \tau_1$$

The **equivalent inertia** J_e and **equivalent damping coefficient** B_e in the output are:

$$J_e = J_1 \left(\frac{N_2}{N_1} \right)^2 + J_2 = J_1 N^2 + J_2 \quad \text{and} \quad B_e = B_1 \left(\frac{N_2}{N_1} \right)^2 + B_2 = B_1 N^2 + B_2$$

The **equation of motion** is:

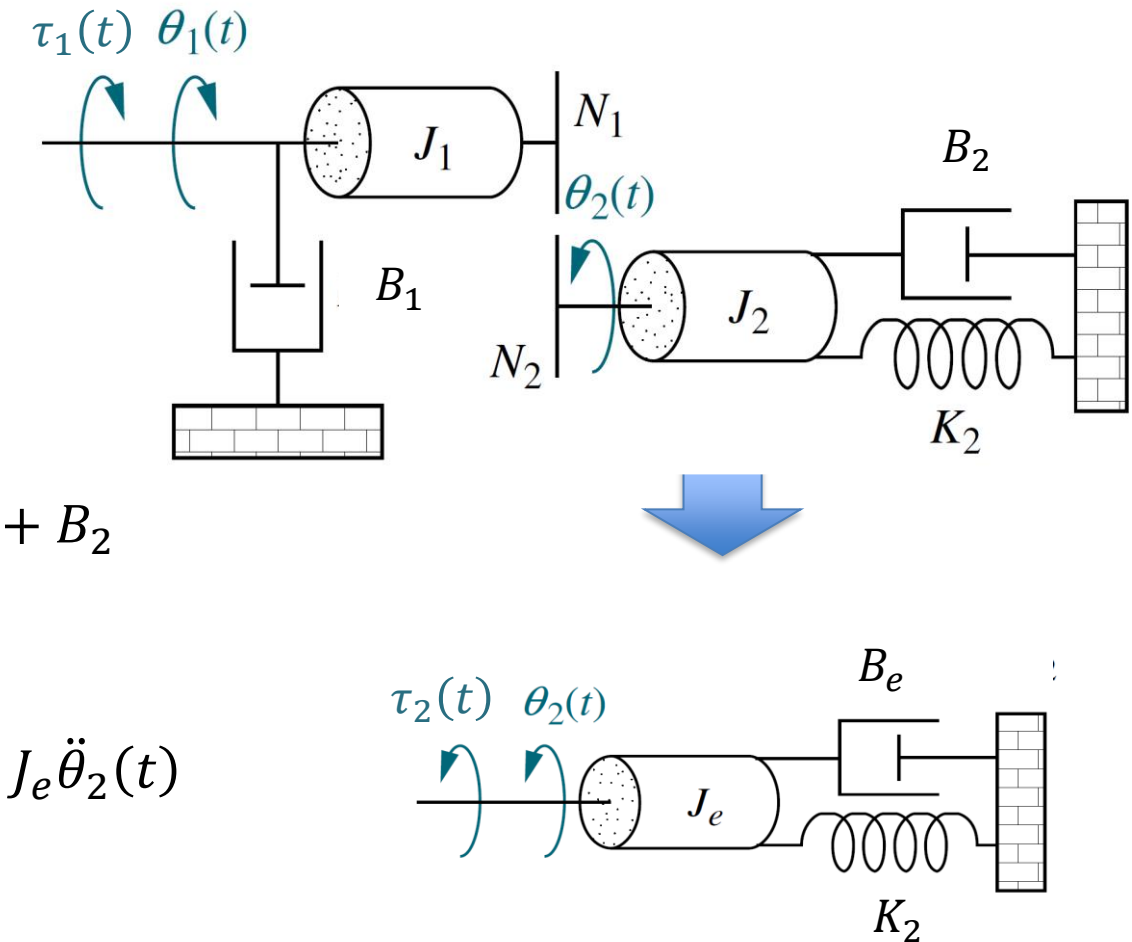
$$\tau_2(t) - K_2 \theta_2(t) - B_e \dot{\theta}_2(t) = J_e \ddot{\theta}_2(t) \rightarrow N \tau_1(t) - K_2 \theta_2(t) - B_e \dot{\theta}_2(t) = J_e \ddot{\theta}_2(t)$$

Take **Laplace** transform with zero initial conditions to find the transfer function:

$$N T_1(s) = K_2 \theta_2(s) + B_e s \theta_2(s) + J_e s^2 \theta_2(s) = (K_2 + B_e s + J_e s^2) \theta_2(s)$$

$$\frac{\theta_2(s)}{T_1(s)} = \frac{N}{J_e s^2 + B_e s + K_2}$$

Transfer Function Model



$$\frac{\tau_2}{\tau_1} = \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} = \frac{N_2}{N_1} = N$$

Modeling of Rotational Mechanical Systems

Example 5

The following system shows gears driving a rotational inertia, spring, and viscous damper system. Find the transfer function, $\theta_1(s)/T_1(s)$ for the following system.

First, find the equivalent **gearless** system.

In this system, all gears have **inertia** and for some shafts there is **viscous friction**.

We have to reflect the impedances to the **input shaft**.

$$J_e = J_1 + (J_2 + J_3) \left(\frac{N_1}{N_2} \right)^2 + (J_4 + J_5) \left(\frac{N_3}{N_4} \right)^2 \left(\frac{N_1}{N_2} \right)^2$$

$$B_e = B_1 + B_2 \left(\frac{N_1}{N_2} \right)^2$$

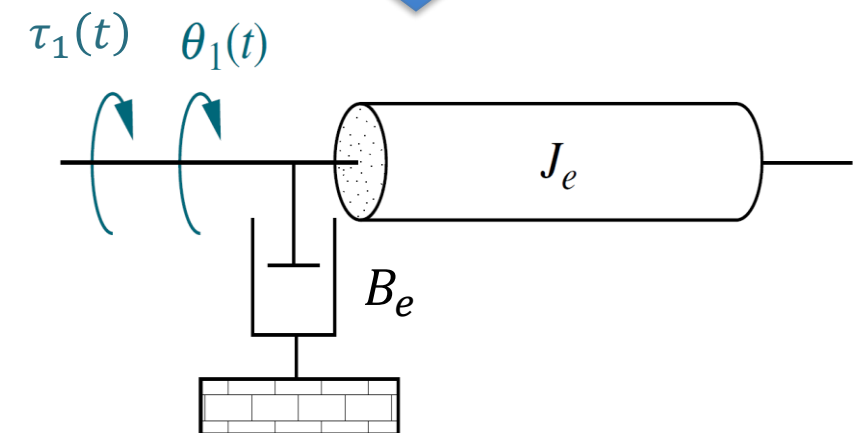
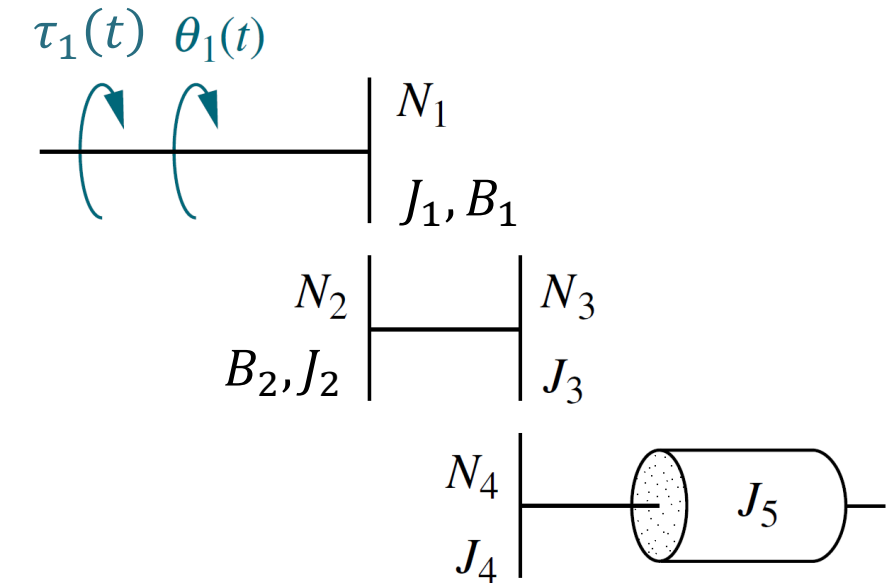
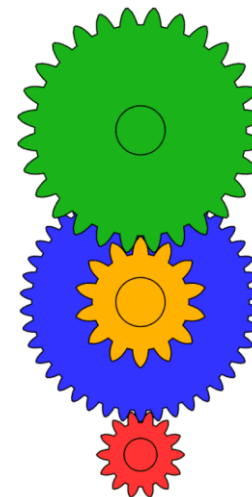
The **equation of motion** is $\rightarrow \tau_1(t) - B_e \dot{\theta}_1(t) = J_e \ddot{\theta}_1(t)$

Take **Laplace** transform to find the transfer function:

$$T_1(s) = B_e s \theta_1(s) + J_e s^2 \theta_1(s)$$

$$\frac{\theta_1(s)}{T_1(s)} = \frac{1}{J_e s^2 + B_e s}$$

Transfer Function Model



THANK YOU